Energy Management Platform
Engineering Handbook

Version 0.1 (Living Document)

This handbook consolidates the architecture, conventions, roadmap and engineering principles of the Industrial IoT SaaS platform being developed for textile processing plants. It is intended to allow another engineer or AI coding agent to understand, extend and maintain the platform with minimal onboarding.

Table of Contents

Project Overview

The platform is an Industrial IoT SaaS system for textile factories. It supports multi-company operation, telemetry collection, dashboards, predictive maintenance, energy management and future AI modules such as OCR-based dyehouse recipe extraction and costing.

Current Technical Stack

FastAPI, SQLAlchemy ORM, PostgreSQL, Docker, Raspberry Pi edge devices, Modbus RTU, REST APIs, Git/GitHub, Docker Compose. Planned additions include InfluxDB, Redis, Grafana, Celery, Flutter and Firebase.

Core Design Principles

• Multi-tenant by design. • Modular architecture. • Strong documentation. • Production-ready code. • Backward compatibility. • Commented source code. • API-first development. • Git-controlled workflow.

Roadmap Snapshot

Completed: Authentication, Companies, Users, Departments, Machines, Machine Components, Telemetry foundation, GitHub integration, documentation. Planned: telemetry ingestion, alarms, dashboards, analytics, mobile apps, predictive maintenance, OCR recipe system and enterprise deployment.

Engineering Handbook Strategy

Each future volume will be expanded directly from the source code so that this handbook becomes the authoritative specification for both human engineers and AI coding assistants.